NISHANT MISHRA

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EDUCATION & EXPERIENCE

University of Michigan, Ann Arbor
Ph.D. in Astronomy & Astrophysics — May 2024 - May 2028
M.S. in Astronomy & Astrophysics — Aug 2022 - May 2024
Advisor(s): Sean Johnson (primary)
Thesis committee: Eric Bell, Jinyi Yang, Camille Avestruz (Department of Physics, UM), Gwen Rudie (Carnegie Observatories)

The Observatories of the Carnegie Institution for Science, Pasadena
Visiting Researcher — Jan 2026 - June 2026
Advisor: Gwen Rudie

University of California, Berkeley
B.A. in Physics, Astrophysics (Honors) — Aug 2017 - May 2021
Advisor(s): Uroš Seljak (thesis), Emmanuel Schaan, Martin White

PUBLICATIONS

First & Second Author (?)

0. **Mishra, N.**, et al., *CUBS+MUSEQuBES: The chemical enrichment properties of the cool circumgalactic medium around star-forming dwarf galaxies at* $z = 0.05 - 0.8$, ApJL, in prep.
-1. **Mishra, N.**, et al., *Differential metal enrichment of neutral, dusty gas in circumgalactic medium around dwarf galaxies traced by damped Lyman-*α *absorption at* $z < 1$, ApJ, in prep.
-2. **Mishra, N.**, Avestruz, C., Johnson, S. D., Bell, E. F., & Liu, Z. 2026, *Tracing the environmental quenching of dwarf galaxies in TNG50*, ApJ, submitted
-3. **Mishra, N.**, Johnson, S. D., Rudie, G. C., et al. 2026, *The Cosmic Ultraviolet Baryon Survey (CUBS) XI: Enriched multi-phase gas beyond the virial radius of a* $\log M_\star/M_\odot \approx 8.4$ *dwarf galaxy at* $z = 0.57$, ApJ, submitted
-4. Johnson, S. D., **Mishra, N.**, Muzahid, S., et al. 2026, *MUSEQuBES: Physical conditions, origins, and multielement abundances of the circumgalactic medium of an isolated, star-forming dwarf galaxy at* $z = 0.57$, ApJL, 996, L30, arXiv:2510.06310
-5. **Mishra, N.**, Johnson, S. D., Rudie, G. C., et al. 2024, *The Cosmic Ultraviolet Baryon Survey (CUBS). IX. The enriched circumgalactic and intergalactic medium around star-forming field dwarf galaxies traced by* O VI *absorption*, ApJ, 976, 149, arXiv:2408.11151
-6. **Mishra, N.**, & Gnedin, N. Y. 2022, *Cosmic Reionization on Computers: Evolution of the flux power spectrum*, ApJ, 928, 174, arXiv:2109.13252
-7. **Mishra, N.**, & Schaan, E. 2019, *Bias to CMB lensing from lensed foregrounds*, PhRvD, 100, 123504, arXiv:1908.08057

Co-Author (?)

0. Kakoly, A., et al. (incl. **Mishra, N.**), *The circumgalactic medium of* $L_\star$ *galaxies at* $z \approx 1.5$ *traced by DESI: Evidence that a quasi-static hot phase has not yet formed*, MNRAS, in prep.
-1. Kumar, S., Chen, H.-W., Rudie, G. C., et al. (incl. **Mishra, N.**) 2026, *The Cosmic Ultraviolet Baryon Survey X: Density and pressure profiles of clumpy circumgalactic gas at* $z = 0.4 - 0.7$, ApJ, accepted, arXiv:2610.36088

-2. Piacitelli, D. R., Brooks, A. M., Sanchez, N. N., et al. (incl. **Mishra, N.**) 2026, *The Simulated Oxygen Shortage (SOS): Mapping the missing* O VI *in simulated dwarf galaxies to subgrid physics*, ApJ, submitted, arXiv:2605.15275

-3. Liu, Z., Johnson, S. D., Bell, E. F., et al. (incl. **Mishra, N.**) 2026, *A comprehensive study of morphology and kinematics in extended nebulae around UV-luminous quasars at* $z \approx 1$, ApJ, 1008, 247, arXiv:2605.01981

-4. Fuller, E., Johnson, S. D., Stern, J., et al. (incl. **Mishra, N.**) 2026, *Under pressure: Ultraviolet emission-line ratios as barometers of active galactic nuclei feedback mechanisms*, ApJ, 1000, 229, arXiv:2602.17039

-5. Liu, Z., Johnson, S. D., Li, J. I.-H., et al. (incl. **Mishra, N.**) 2025, *The morphology and kinematics of a giant, symmetric nebula around a radio-loud quasar 3C 57: Extended rotating gas or biconical outflows?*, ApJ, 984, 140, arXiv:2503.12597

-6. Li, J. I.-H., Johnson, S. D., Avestruz, C., et al. (incl. **Mishra, N.**) 2024, *Fast and flexible inference framework for continuum reverberation mapping using simulation-based inference with deep learning*, ApJ, 977, 223, arXiv:2407.14621

-7. Johnson, S. D., Liu, Z., Li, J. I.-H., et al. (incl. **Mishra, N.**) 2024, *Discovery of optically emitting circumgalactic nebulae around the majority of UV-luminous quasars at intermediate redshift*, ApJ, 966, 218, arXiv:2404.00088

-8. Li, J. I.-H., Johnson, S. D., Boettcher, E., et al. (incl. **Mishra, N.**) 2024, *The Cosmic Ultraviolet Baryon Survey (CUBS). VIII. Group environment of the most luminous quasars at* $z \approx 1$, ApJ, 965, 143, arXiv:2403.03983

-9. Schutt, T., Maniyar, A. S., Schaan, E., Coulton, W. R., & **Mishra, N.** 2024, *New temperature inversion estimator to detect CMB patchy screening by large-scale structure*, PhRvD, 109, 103539, Erratum: PhRvD, 112, 069901 (2025), arXiv:2401.13040

-10. Liu, Z., Johnson, S. D., Li, J. I.-H., et al. (incl. **Mishra, N.**) 2024, *The first comprehensive study of a giant nebula around a radio-quiet quasar in the* $z < 1$ *Universe*, MNRAS, 527, 5429, arXiv:2309.00053

-11. Ren, R., Bathurst, C., Chang, Y. Y., et al. (incl. **Mishra, N.**) 2021, *Design and characterization of a phonon-mediated cryogenic particle detector with an eV-scale threshold and 100 keV-scale dynamic range*, PhRvD, 104, 032010, arXiv:2012.12430

-12. Palliyaguru, N. T., Corsi, A., Kasliwal, M. M., et al. (incl. **Mishra, N.**) 2016, *Radio follow-up of gravitational-wave triggers during Advanced LIGO O1*, ApJL, 829, L28, arXiv:1608.06518

PRESENTATIONS

Invited Talks

· *Borthakur Group Meeting*, Arizona State University, Tempe, AZ, June 2026.
· *Riverside Astrophysics Seminar*, UC Riverside, Riverside, CA, May 2026.
· *Oh Group Meeting*, UC Santa Barbara, Santa Barbara, CA, March 2026.
· *Yale Galaxy Lunch*, Yale University, New Haven, CT, December 2025.

Contributed Talks

· *GalScale: The many scales of galaxy environments*, Ascona, Switzerland, July 2026.
· *MMC2025: Multi-phase, Multi-temperature, and Complex*, Olbia, Sardinia, Italy, October 2025.
· *Cosmic Ecosystems*, Perimeter Institute for Theoretical Physics, Waterloo, ON, Canada, July 2025, Slides, Video.
· *Galactic Frontiers II*, Dartmouth College, Hanover, NH, June 2025, Slides.
· *5th Magellan Science Meeting*, Carnegie Science Earth and Planets Laboratory, Washington, DC, May 2025.
· *Baryons Beyond Galactic Boundaries*, IUCAA, Pune, India, December 2024.

- *CGM-Chile 2024: Resolving the Circumgalactic Medium and its Impact on Galaxy Evolution*, ESO, Santa Cruz, Colchagua, Chile, November 2024, Slides.
- *Multiphase Madness*, Center for Astrophysics, Harvard & Smithsonian, Cambridge, MA, August 2024, Video.
- *Simons Observatory Collaboration*, September 2019, Slides.

Flash Talks/Posters

- *The Metal-Poor Frontier: Understanding Low-Metallicity Stars and Galaxies Across Cosmic Time*, Space Telescope Science Institute, Baltimore, MD, May 2026, Video.
- *BerkeleyLab Undergraduate Researchers (BLUR) Poster Session*, Lawrence Berkeley National Laboratory, Berkeley, CA, May 2022, Poster.
- *Science Undergraduate Laboratory Internship (SULI) Poster Session*, Fermi National Accelerator Laboratory, Batavia, IL, December 2021, Poster.
- *Berkeley Physics Undergraduate Research Scholars (BPURS) Symposium*, UC Berkeley, Berkeley, CA, May 2021, Poster.
- *Berkeley Physics Undergraduate Poster Session (Cal Week in Physics)*, UC Berkeley, Berkeley, CA, April 2021, Poster.
- *Science Undergraduate Laboratory Internship (SULI) Poster Session*, Fermi National Accelerator Laboratory, Batavia, IL, August 2020, Poster.
- *Undergraduate Physics Symposium*, UC Berkeley, Berkeley, CA, 2018, Poster.

OBSERVING

Observing Time Awarded as Principal Investigator

- Magellan/Baade IMACS: 7.5 nights (2025B, 2026A, 2026B)
- Magellan/Clay MIKE: 2 nights (2026A, 2026B)

Co-Investigator on Awarded Programs

- Fuller, E., et al. (incl. **Mishra, N.**) 2026, *HST snapshot survey for a three-fold increase in UV obscured quasar spectra to constrain drivers of AGN feedback*, HST Proposal. Cycle 34, ID. #18372, SNAP, 100 orbits, STScI
- Johnson, S., Chen, H.-W., Qu, Z., et al. (incl. **Mishra, N.**) 2025, *Baryonic Escape, Abundances, and Thermodynamics in Cosmic Ecosystems (BEATriCE)*, HST Proposal. Cycle 33, ID. #18107, GO, 70 orbits, SciX
- Liu, W., Johnson, S., Boettcher, E., et al. (incl. **Mishra, N.**) 2025, *Confirming and characterizing a fast, cool inflow toward a luminous quasar at* $z = 0.78$, HST Proposal. Cycle 33, ID. #18039, GO, 9 orbits, SciX
- Johnson, S., Boettcher, E., Chen, H.-W., et al. (incl. **Mishra, N.**) 2024, *The first multi-phase survey of the circumgalactic medium of quasar hosts at* $z < 1$, HST Proposal. Cycle 32, ID. #17803, GO, 51 orbits, SciX
- Johnson, S., Gaspari, M., Greene, J. E., et al. (incl. **Mishra, N.**) 2024, *Diagnostics of AGN feedback with the first UV atlas of obscured quasars*, HST Proposal. Cycle 32, ID. #17800, SNAP, 147 orbits, SciX

Total Observing Experience

- Magellan/Baade IMACS: 18.5 nights (incl. 11 on programs led by S. Johnson)
- Magellan/Clay MIKE: 4 nights (incl. 2 on programs led by S. Johnson)

AWARDS & GRANTS

Rackham Graduate School Research Grant 2024, 2026

Rackham Graduate School Travel Grant	2024, 2025, 2026
NSF Graduate Research Fellowship, Honorable Mention	2022, 2024
Berkeley Lab Undergraduate Research (BLUR) Grant	2022
Science Undergraduate Laboratory Internship (SULI), Fermilab	2020, 2021
Berkeley Physics Undergraduate Research Scholar (BPURS), UC Berkeley	2019, 2020, 2021
Clark Scholar	2016

TEACHING

University of Michigan

Graduate Student Instructor, ASTRO 102: Introductory Astronomy	Winter 2023, Fall 2023

UC Berkeley

Student Facilitator, Physics 98 DeCal: Lasers for Everyone	Spring 2021
Undergraduate Student Instructor, Astronomy C10: Introduction to General Astronomy	Fall 2020

SERVICE

University of Michigan

Graduate Student Member, Magellan/MDM Time Allocation Committee	2026A, 2027A
Graduate Student Member, Astronomy Graduate Admissions Committee (Core)	2026–2027
Organizer, Galaxies Group weekly seminar and journal club, Department of Astronomy	2023–2026

OUTREACH

University of Michigan

Founding Member, Senior Center Astronomy Outreach, Department of Astronomy	2024–present
Physical Sciences Division Representative, Rackham Student Government	2025–2026

UC Berkeley

Founder and Coordinator, Undergraduate Research Fair, Department of Physics 2020–2021
Matches undergraduates with faculty research groups; now a recurring department event.

Instructor, Splash high school program: "An Introduction to Dark Matter Physics" 2020